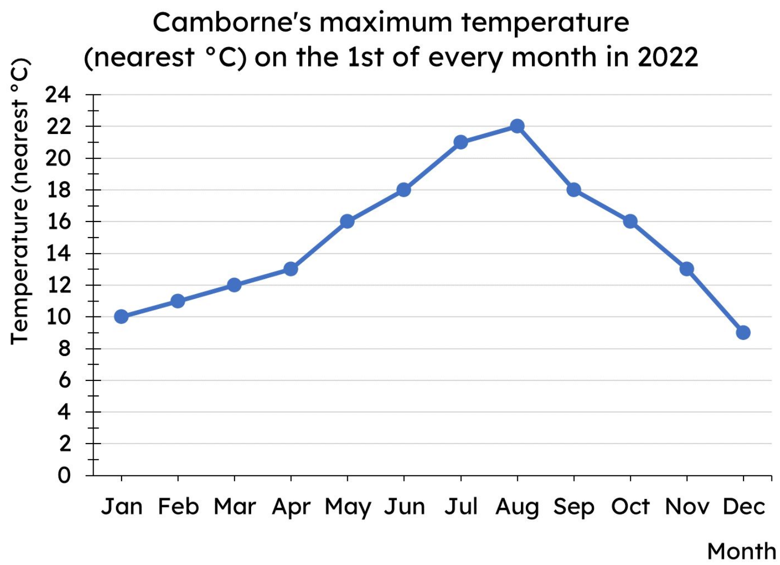
Calculating the mean

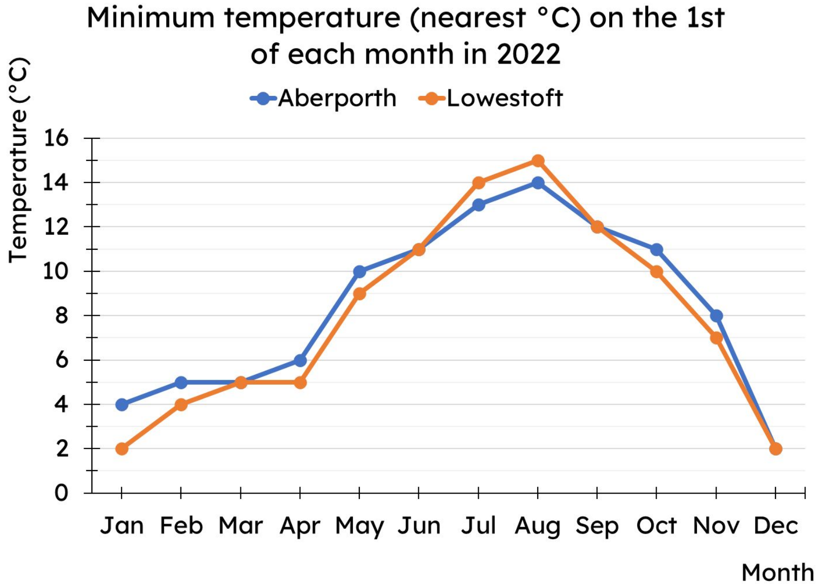
Task A: Calculating the mean from a line graph

1) Use the line graph to complete the table of data.



Jan	10
Feb	11
Mar	12
Apr	13
May	16
Jun	18
Jul	21
Aug	22
Sep	18
Oct	16
Nov	13
Dec	9

2) Calculate the mean of each data set shown on the line graph.



Aberporth

Total temperature = 101
Mean minimum temperature =
101 ÷ 12 = 8.42 °C (2dp)

Lowestoft

Total temperature = 96
Mean minimum temperature =
96 ÷ 12 = 8 °C

Task B: Calculating the mean from an ungrouped frequency table

1) A class take a quiz, marked out of 6.
The frequency table shows the results for the class.

Quiz score	0	1	2	3	4	5	6
Frequency	3	4	4	7	10	4	1

a) How many pupils are in the class?

$3 + 4 + 4 + 7 + 10 + 4 + 1 = 33$ pupils

b) How many marks did the class get as a group?

$0 + 4 + 8 + 21 + 40 + 20 + 6 = 99$ marks

c) Hence, what is the mean marks per pupil?

$99 \div 33 = 3$ marks per pupil

2) The frequency table shows the number of bedrooms in different houses within an area of a town.

Which calculation correctly finds the mean for this frequency table? For each incorrect answer, why is it wrong?

Bedrooms	Frequency
1	20
2	93
3	54
4	33

a) $\frac{500}{200} = 2.5$ ✓

b) $\frac{200}{10} = 20$

c) $\frac{500}{4} = 125$

d) $\frac{500}{10} = 50$

e) $\frac{200}{4} = 50$

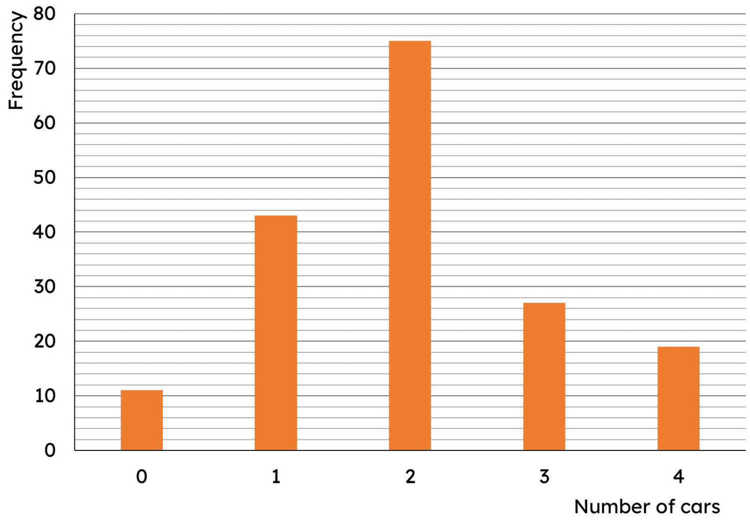
f) $\frac{10}{200} = 0.05$

In b, d and f, the 10 comes from adding up the first column, which is meaningless.

And the 4 in c and e is how many categories, not how many pieces of data.

Task C: Calculating the mean from a bar chart

1) This bar chart shows the number of cars per household.



a) How many households are there?

175 households

b) How many cars are there in total across all households?

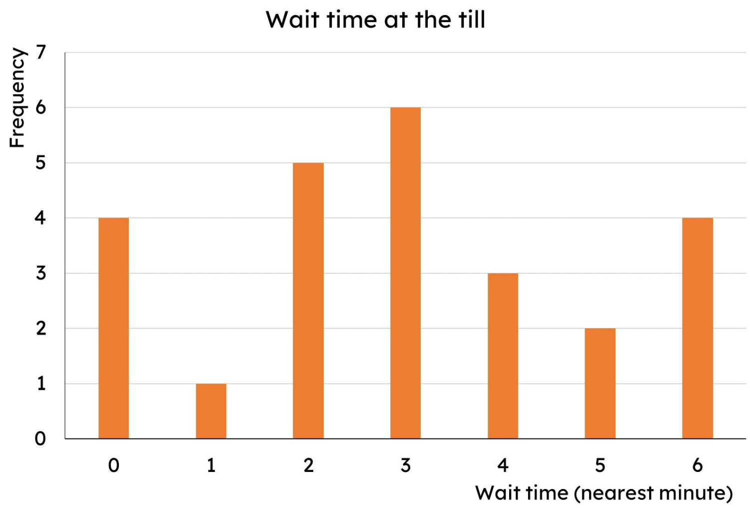
350 cars

c) What is the mean number of cars per household?

$350 \div 175 = 2$ cars per household

2) The bar chart shows the length of time customers spent waiting at a self-service till (to the nearest minute).

What is the mean wait time?



Total time by all customers
$= 0 + 1 + 10 + 18 + 12 + 10 + 24$
$= 75$ minutes

6 lots of 3 minutes

Total customers
$= 4 + 1 + 5 + 6 + 3 + 2 + 4$
$= 25$ customers

Mean wait time
$= 75 \div 25 = 3$ minutes